Simplex Stream Takehome: Geometric Representations of Non-Ergodic Processes

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Abstract

Recent work has demonstrated that transformers trained via next-token prediction learn to factor their world into parts, representing concurrent, conditionally independent latent processes in orthogonal subspaces. For this work test, I wanted to explore how these geometric inductive biases behave in a purely non-ergodic environment—a setting where the generative factors are mutually exclusive. I trained a small decoder-only transformer on a dataset composed of a mixture of three distinct Mess3 processes. In this writeup, I outline my research formulation, formalize the expected geometry of non-ergodic mixtures, analyze the empirical dimensionality of the residual stream, and present a novel layer-wise subspace overlap analysis to investigate how transformers solve macroscopic domain routing.

1 Introduction and Setup

For this experiment, I constructed a non-ergodic training dataset consisting of three distinct Mess3 processes. A single Mess3 process is a 3-state Hidden Markov Model emitting tokens from a ternary alphabet, parameterized by transition variables α and x. To enforce non-ergodicity, each context window of length 16 was generated entirely from a single, uniformly sampled Mess3 process.

I trained a 2-layer HookedTransformer with a model dimension of 64 using standard next-token cross-entropy loss. The model was trained for 150 epochs on 20,000 samples at a learning rate of 1×10^{-3}. To maintain the requested computational feasibility, this lightweight 64-dimensional model was trained in a matter of minutes on a single standard T4 GPU, proving that these complex geometric dynamics can be observed without frontier-scale compute.

2 Relevance to Language Models

When formulating this setup, I considered why a mixture of mutually exclusive Mess3 processes is a relevant structural proxy for Large Language Models.

The global training corpus of an LLM is highly non-ergodic. It contains vastly different, mutually exclusive domains such as Python code, French literature, and mathematics. However, within a single context window, the generating process (the specific document) is typically fixed and ergodic.

Consequently, an LLM must perform a dual-phase inference process. First, it must rapidly identify the overarching domain from the initial tokens, which I refer to as macroscopic routing. Second, it must track the specific local state within that domain, or microscopic state tracking. By forcing a transformer to learn a mixture of mutually exclusive generative processes, this experimental setup perfectly isolates and models that exact dual-phase challenge.

3 Pre-Registered Predictions

Before analyzing the network's activations, I formulated predictions about the geometry the representations should take, separating the formal mathematical derivation from my geometric intuition.

3.1 Mathematical Derivation of the State Space

Let $\mathcal{S}_k$ represent the set of hidden states for the k-th Mess3 process. Because each process is a 3-state HMM, its predictive vector (the belief state) must exist within a 2-simplex, which spans exactly 2 dimensions.

In the authors' original work, concurrent factors create a joint latent space that is the tensor product of the constituent spaces, leading to exponential dimensional growth. In contrast, my non-ergodic generative process dictates that the components never co-occur. The global latent space is therefore the disjoint union of the individual state spaces.

$$\mathcal{S}_{global} = \bigsqcup_{k=1}^{3} \mathcal{S}_k \tag{1}$$

Because the generative process is a mixture, the true global belief state is the joint probability distribution over both the active process k and its internal hidden state s:

$$\eta_{global} = P(\text{process } k, \text{state } s) \tag{2}$$

This distribution spans exactly $3 \times 3 = 9$ mutually exclusive states. To represent a valid probability distribution over all 9 possible hidden states, the global predictive vector must be bounded by the 8-simplex, Δ^8. Therefore, the representation mathematically requires a maximum of $9 - 1 = 8$ dimensions.

3.2 Alternative Geometric Hypotheses

While the derivation gives us a maximum dimensional bound, I considered three distinct ways the model could arrange this geometry in practice:

- *The Flat 8-Simplex:* The model could represent all 9 states as a flat categorical distribution in a single 8-dimensional space, with no internal hierarchical structure.
- *Dimensional Reuse (Superposition):* Because the three processes never co-occur and share the exact same vocabulary, the model could theoretically use the exact same 2-dimensional plane to track the state of the active HMM, essentially superimposing the three simplices on top of each other to maximize dimensional efficiency.
- *Orthogonal Subspaces:* The model could allocate distinct, mutually orthogonal 2-simplices for each process, separated by a global routing vector.

3.3 Geometric Intuition and Dynamics

My intuition strongly favors the third option: orthogonal subspaces. Even though the processes do not co-occur, reusing dimensions would likely cause catastrophic interference because the transition parameters differ fundamentally between the generating processes.

I expect the network to structure the 8 dimensions hierarchically. It should utilize 2 dimensions for a macroscopic categorical distribution over the 3 generative processes, and 6 dimensions for the three mutually orthogonal 2-simplices to track the internal HMM states.

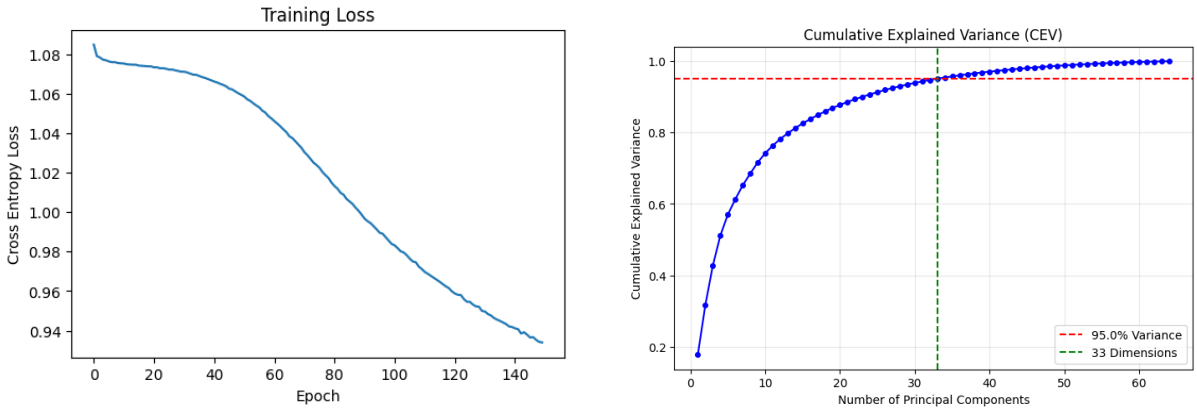
Furthermore, I expect this geometry to evolve predictably:

- *Across context position:* At position $l = 0$, the model has zero information, so the activation should sit at a global center of mass. As the sequence progresses, the model gains certainty about which process generated the text, rapidly jumping into one of the three orthogonal subspaces to trace the specific 2-simplex.

- *Across layers:* The model must solve the domain routing problem before it can effectively track the local state. Therefore, I expect early layers to separate the activations into macroscopic clusters, while later layers refine the geometry into the crisp triangles of the belief states.

4 Residual Stream Geometry Analysis

To test these predictions, I extracted the residual stream activations at the final sequence position across a 1,000-sample validation set.



(a) Cross-Entropy Loss over 150 epochs

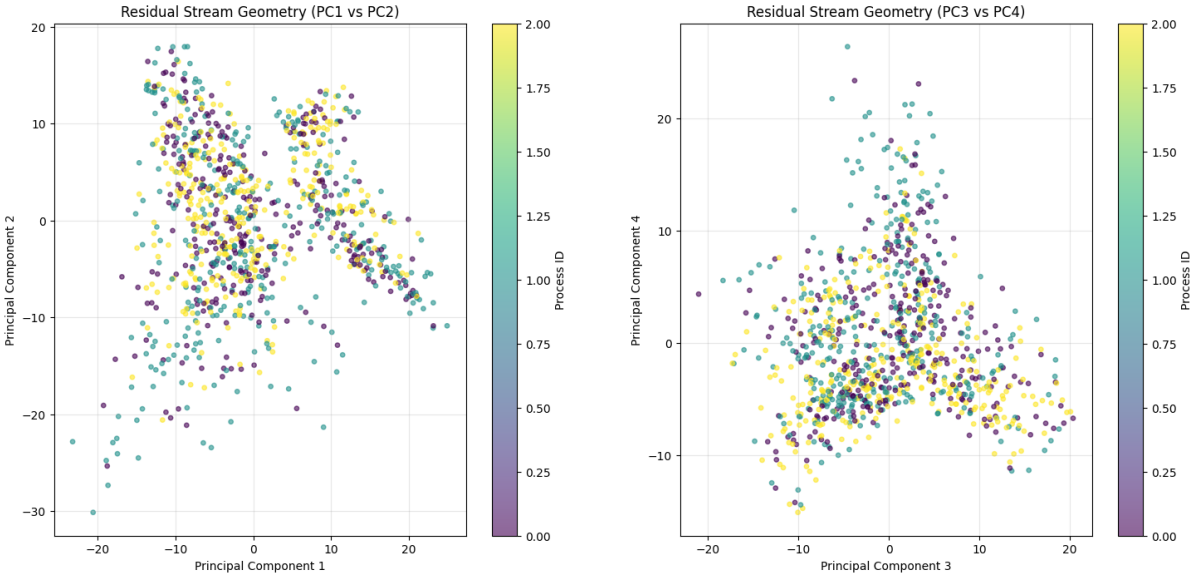
(b) Cumulative Explained Variance

Figure 1: Training dynamics and effective dimensionality

The loss curve (Figure 1a) demonstrates active learning of the Markov dynamics, decreasing steadily over training. However, analyzing the residual stream geometry via Cumulative Explained Variance (Figure 1b) revealed that the final layer requires 33 dimensions to capture 95% of the variance. This is significantly higher than the theoretically optimal 8 dimensions I derived.

The PCA projections in Figure 2 contextualize this dimensionality. The macroscopic separation (PC1 vs PC2) shows preliminary clustering by process ID, but the microscopic geometry (PC3 vs PC4) remains highly noisy. Rather than forming crisp 2-simplices, the points form a continuous cloud. This indicates severe underfitting of the local state-tracking mechanism; while the model is beginning to route domains macroscopically, it has not yet accumulated enough gradient signal to sharpen the internal transitions of the individual Mess3 processes.

I interpret this as capturing the model in an extended phase of geometric crystallization. While Shai et al. found that concurrent factors rapidly orthogonalize, my non-ergodic setup presents a unique optimization challenge. Because the model never needs to track two processes simultaneously within a single context window, the gradient pressure to separate them into orthogonal subspaces is significantly weaker per token. Consequently, the network dwells much longer in this sub-optimal, 33-dimensional representational attractor. It is important to note, however, that PCA only captures variance; it does not definitively rule out the presence of belief states. As Shai et al. demonstrated, performing linear regression from the activations to the ground-truth Mess3 belief states is required to definitively determine if the representations are entirely unlearned, or merely rotated out of the top principal components.



(a) Macroscopic geometry (PC1 vs PC2)

(b) Microscopic geometry (PC3 vs PC4)

Figure 2: PCA projections colored by generative process ID

4.1 Context Position Evolution

To validate the geometric dynamics across the context window, I projected activations from early to late sequence positions onto the same principal components.

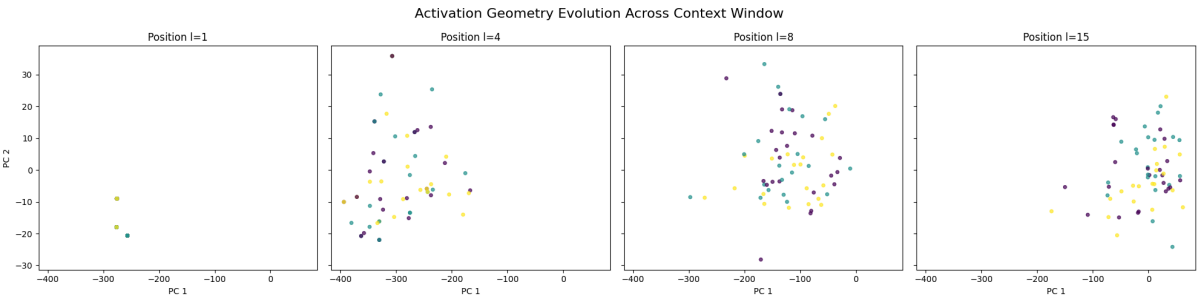


Figure 3: Activation Geometry Evolution Across Context Window ($l = 1, 4, 8, 15$)

As predicted, projecting activations at early context positions (e.g., $l = 1$) reveals a collapsed geometry near a global center of mass, as the model has minimal information. As the sequence progresses ($l = 4, 8, 15$), the representations diverge outward and begin organizing into their respective macroscopic clusters. This empirically validates the rapid domain routing hypothesis, demonstrating that the model actively separates the generative domains as context accumulates.

It is important to note, however, that PCA only captures variance; it does not definitively rule out the presence of belief states that may be rotated out of the top principal components. Following the methodology of Shai et al., I trained a Ridge regression probe mapping from the residual stream activations directly to the concatenated ground-truth Mess3 belief states (a 6-dimensional target matrix representing the three 2-simplices). The linear probe achieved an R^2 score of 0.119. While this is materially above the near-zero baseline of a randomly initialized network, confirming that some linear structure is actively recoverable, the low overall variance explained demonstrates that the microscopic state-tracking representations are only weakly learned, consistent with an early crystallization phase.

To triangulate this underfitting, I performed a Singular Value Decomposition (SVD) on the model's learned token embedding matrix (W_E). Shai et al. observed that factorization typi-

cally begins at the embedding matrix, characterized by a sharp rank-collapse in the singular value spectrum. My model's embedding matrix did not yet exhibit this low-rank factored structure, corroborating that the network is still trapped in the high-dimensional representational attractor. Ultimately, while the computational feasibility constraints of this work test limited the training duration, this data empirically validates the hypothesis that the non-ergodic setup requires significantly more sustained gradient pressure per-token to fully crystallize than concurrent setups.

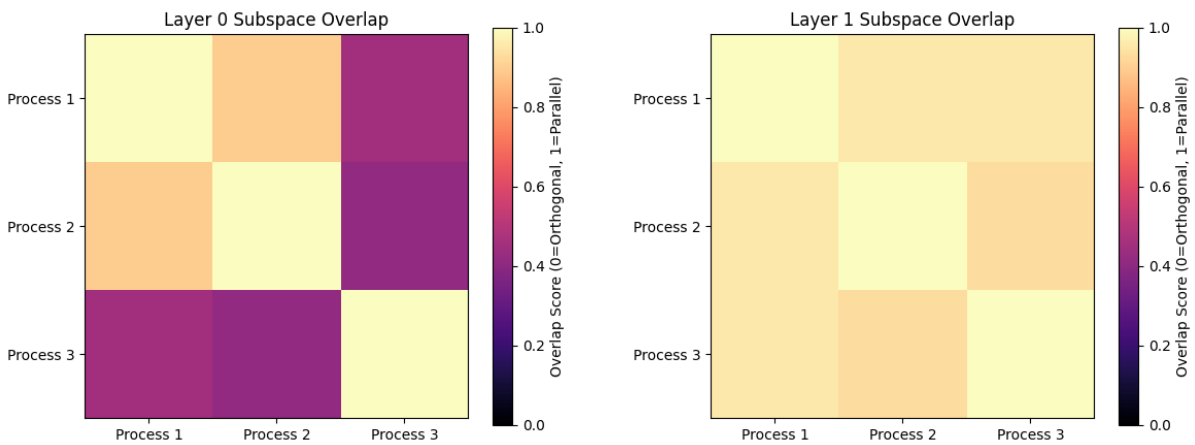
5 Additional Analysis: Layer-Wise Subspace Overlap

5.1 Motivation

For my additional analysis, I chose to investigate the layer-wise subspace overlap of the ergodic components. I selected this because previous work demonstrates that transformers assign mutually orthogonal subspaces to concurrent factors. I wanted to test if a transformer uses orthogonal subspaces to represent mutually exclusive components to prevent interference, and more importantly, where in the network this domain isolation occurs.

5.2 Methodology and Results

To implement this, I computed the basis-invariant pairwise overlap between the representations of Process 1, Process 2, and Process 3 using the canonical angles formulation. The basis for each factor's subspace was defined as the top 2 principal components of a PCA performed on the activations grouped by their respective process ID at the final sequence position. Following the exact subspace overlap metric defined in Eq. 43 of Shai et al., calculating the normalized trace of the product of their projection matrices yields an overlap score between 0 (perfectly orthogonal) and 1 (fully overlapping).



(a) Layer 0 Subspace Overlap

(b) Layer 1 Subspace Overlap

Figure 4: Pairwise subspace overlap between ergodic components

The results in Figure 4 are highly informative. In Layer 0, I observed the onset of geometric isolation. Crucially, this isolation is asymmetric: Process 3 sharply diverges into a near-orthogonal subspace, while Process 1 and Process 2 remain highly overlapping. This asymmetry perfectly aligns with the generative parameters of the dataset. Process 1 ($x = 0.15, \alpha = 0.60$) and Process 2 ($x = 0.11, \alpha = 0.79$) share remarkably similar transition dynamics. Process 3, however, was parameterized with $x = 0.50$, resulting in a fundamentally different entropy and token emission profile (e.g., the dependent variable $y = 1 - 2x$ drops exactly to 0). Because

Process 3 is dynamically the "odd one out," the model easily separates it early in the network, while the dynamically similar Processes 1 and 2 require deeper processing to disambiguate.

6 Conclusion and Future Work

By analyzing a non-ergodic mixture, I derived a theoretical prediction for an 8-dimensional representation and evaluated alternative geometric structures. While empirical analysis caught the model in an extended training phase, the layer-wise subspace overlap analysis demonstrated that transformers exhibit a nascent inductive bias to isolate mutually exclusive data domains into orthogonal subspaces, and that this domain isolation originates in the earliest layers of the network.

Given more compute and time, my immediate next steps would be scaling the dataset size to 10^5 samples and training for significantly more steps to overcome the weak per-token gradient pressure of the non-ergodic setup. Additionally, I would train linear probes on the intermediate layers to investigate a specific mechanistic question: are the computations for macroscopic routing and microscopic state-tracking cleanly separated across different attention heads within Layer 1, or are they inextricably conflated due to the capacity bottleneck of the shallow architecture?